

LAB SHEET - 4

Lab Assignment: SQL User Management & Access Control

```
mysql> CREATE DATABASE lab_db;
Query OK, 1 row affected (0.03 sec)

mysql> USE lab_db;
Database changed
mysql> CREATE TABLE Student (
  ->     id INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     marks INT
  -> );
Query OK, 0 rows affected (0.13 sec)

mysql> CREATE TABLE Course (
  ->     course_id INT PRIMARY KEY,
  ->     course_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)
```

Tasks

Part A: User Creation

1. Create a new user named `user1` with a password.
2. Create another user `user2` with limited access.

```
mysql> CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password1';
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password2';
Query OK, 0 rows affected (0.01 sec)
```

Part B: Granting Privileges

3. Grant **SELECT** and **INSERT** privileges on a table `Student` to `user1`.
4. Grant **ALL** privileges on a table `Course` to `user2`.
5. Provide **WITH GRANT OPTION** to `user1` and observe its effect.

```
mysql> GRANT SELECT, INSERT ON lab_db.Student TO 'user1'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT ALL PRIVILEGES ON lab_db.Course TO 'user2'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT SELECT, INSERT ON lab_db.Student TO 'user1'@'localhost' WITH GRANT OPTION;
Query OK, 0 rows affected (0.01 sec)
```

Part C: Revoking Privileges

6. Revoke **INSERT permission** from user1 on the Student table.
7. Revoke all privileges from user2 on the Course table.

```
mysql> REVOKE INSERT ON lab_db.Student FROM 'user1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE ALL PRIVILEGES ON lab_db.Course FROM 'user2'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

Part D: User Deletion

8. Drop/Delete user user2 from the database.
9. Attempt to login using deleted credentials and observe the result.

```
mysql> DROP USER 'user2'@'localhost';
Query OK, 0 rows affected (0.03 sec)
```

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

```
mysql> SELECT user, host FROM mysql.user;
+-----+-----+
| user          | host          |
+-----+-----+
| dept_user     | localhost    |
| mysql.infoschema | localhost    |
| mysql.session | localhost    |
| mysql.sys     | localhost    |
| root          | localhost    |
| user1         | localhost    |
+-----+-----+
6 rows in set (0.02 sec)
```

```
mysql> SHOW GRANTS FOR 'user1'@'localhost';
+-----+-----+
| Grants for user1@localhost |
+-----+-----+
| GRANT USAGE ON *.* TO `user1`@`localhost` |
| GRANT SELECT ON `lab_db`.`student` TO `user1`@`localhost` WITH GRANT OPTION |
+-----+-----+
2 rows in set (0.01 sec)
```